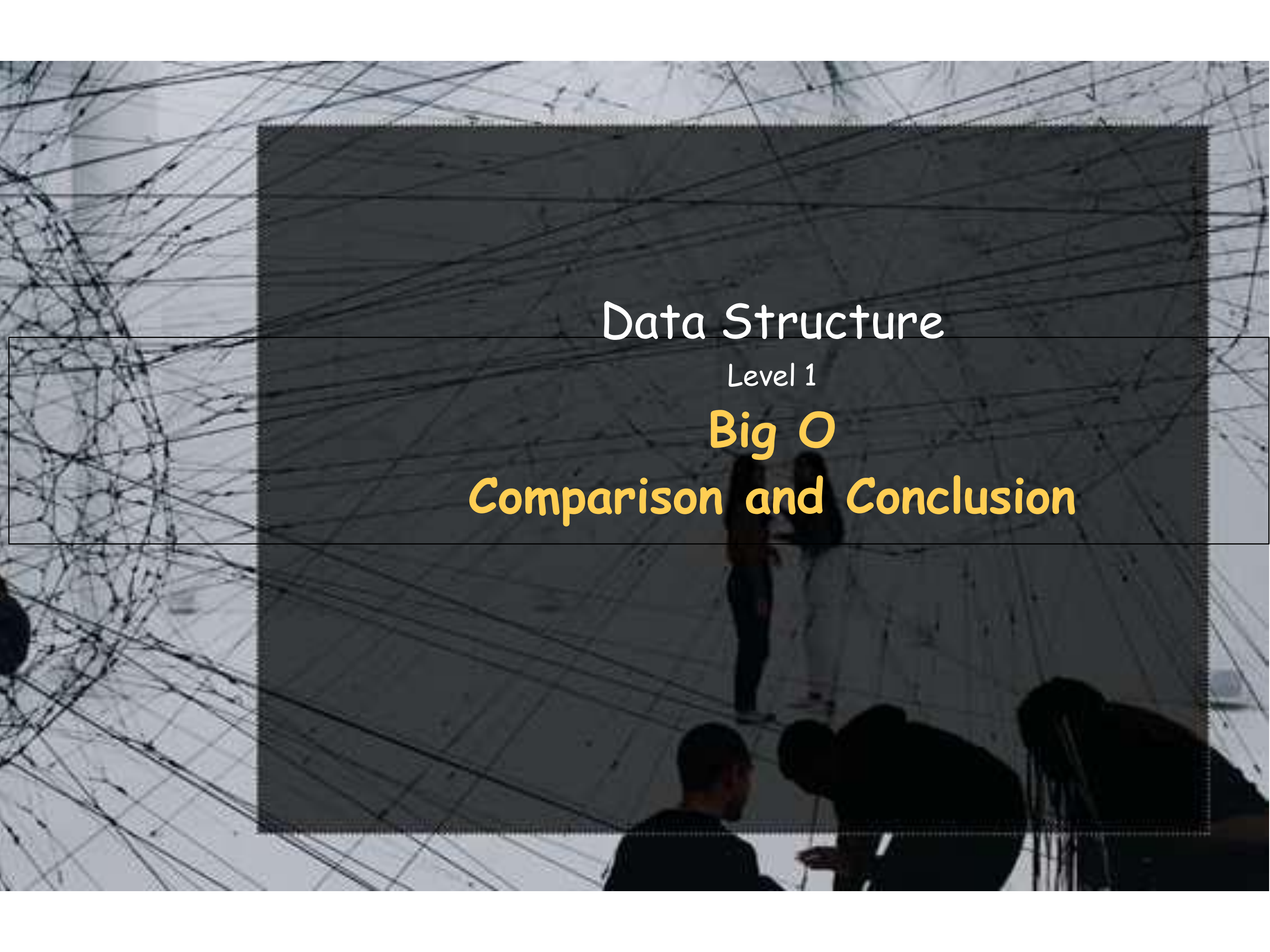


Data Structure Level 1

Big O Comparison and Conclusion



Data Structure

Level 1

Big O

Comparison and Conclusion

Big O Cheat Chart



Growth Rate comparison

	n=1	n=2	n=4	n=8	n=16	n=32
1	1	1	1	1	1	1
$\log n$	0	1	2	3	4	5
n	1	2	4	8	16	32
$n \log n$	0	2	8	24	64	160
n^2	1	4	16	64	256	1024
n^3	1	8	64	512	4096	32768
2^n	2	4	16	256	65536	4294967296
$n!$	1	2	24	40320	20.9T	Don't ask!

Time



$O(n!)$: Exhaustive Search

$O(2^n)$: Exponential

$O(n^3)$: Cubic

$O(n^2)$: Quadratic

$O(n \log n)$: log linear

$O(n)$: linear

$O(\log n)$: logarithmic

$O(1)$: Constant

Always take the worst Big O

$$F(t) = 10 + O(n) + O(\log n) + O(n^2) + O(n^3)$$

$$F(t) = O(n^3)$$

Thank you 😊